

Math 162

Midterm Exam 2 solutions

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1. (a) We observe that

$$\frac{d}{dx} \sinh x = \frac{d}{dx} \left[\frac{1}{2}(e^x - e^{-x}) \right] = \frac{1}{2}(e^x + e^{-x}) = \cosh x$$

and

$$\frac{d}{dx} \cosh x = \frac{d}{dx} \left[\frac{1}{2}(e^x + e^{-x}) \right] = \frac{1}{2}(e^x - e^{-x}) = \sinh x.$$

The Pythagorean-type identity is verified by straightforward algebra:

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left(\frac{1}{2}(e^x + e^{-x}) \right)^2 - \left(\frac{1}{2}(e^x - e^{-x}) \right)^2 \\ &= \frac{1}{4}(e^{2x} + 2 + e^{-2x}) - \frac{1}{4}(e^{2x} - 2 + e^{-2x}) = 1. \end{aligned}$$

- (b) We first show that both sides of the identity have the same derivative. Observe that

$$\frac{d}{dx} \left[\frac{1}{2}(\cosh x \sinh x - x) \right] = \frac{1}{2}(\sinh^2 x + \cosh^2 x - 1) = \sinh^2 x,$$

where we used both parts of (a). On the other hand, we have

$$\frac{d}{dx} \int_1^{\cosh x} \sqrt{t^2 - 1} dt = \sqrt{\cosh^2 x - 1} \cdot \sinh x = \sinh^2 x.$$

Here the first step used the first fundamental theorem of calculus and the chain rule, and in the second step we have

$$\sqrt{\cosh^2 x - 1} = \sqrt{\sinh^2 x} = |\sinh x| = \sinh x,$$

because $x \geq 0$ implies that $e^x \geq e^{-x}$ and hence $\sinh x \geq 0$.

Since both sides of the identity have the same derivative, it follows that

$$\int_1^{\cosh x} \sqrt{t^2 - 1} dt = \frac{1}{2}(\cosh x \sinh x - x) + C$$

for some $C \in \mathbb{R}$. If we substitute $x = 0$, then the left side evaluates to

$$\int_1^{\cosh 0} \sqrt{t^2 - 1} dt = \int_1^1 \sqrt{t^2 - 1} dt = 0.$$

On the other hand, we have

$$\frac{1}{2}(\cosh 0 \sinh 0 - 0) = \frac{1}{2} \cdot 1 \cdot 0 = 0.$$

It follows that $C = 0$, which proves the desired identity.

2. (a) Apply the definition of the derivative:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{\sin h}{h} - 1}{h} = \lim_{h \rightarrow 0} \frac{\sin h - h}{h^2}.$$

The numerator and denominator both approach 0 as $h \rightarrow 0$, so L'Hôpital's rule applies and we obtain

$$\lim_{h \rightarrow 0} \frac{\sin h - h}{h^2} = \lim_{h \rightarrow 0} \frac{\cos h - 1}{2h}.$$

Again we get something of the form $\frac{0}{0}$, so L'Hôpital yields

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{2h} = \lim_{h \rightarrow 0} \frac{-\sin h}{2} = 0.$$

Thus $f'(0) = 0$.

(b) The definition of the derivative says that

$$f''(0) = \lim_{h \rightarrow 0} \frac{f'(h) - f'(0)}{h} = \lim_{h \rightarrow 0} \frac{f'(h)}{h},$$

where we used $f'(0) = 0$ from part (a). By the quotient rule, we have

$$f'(x) = \frac{x \cos x - \sin x}{x^2}$$

for $x \neq 0$. Thus

$$\lim_{h \rightarrow 0} \frac{f'(h)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h \cos h - \sin h}{h^2}}{h} = \lim_{h \rightarrow 0} \frac{h \cos h - \sin h}{h^3}.$$

This has the form $\frac{0}{0}$, so by L'Hôpital

$$\lim_{h \rightarrow 0} \frac{h \cos h - \sin h}{h^3} = \lim_{h \rightarrow 0} \frac{\cos h - h \sin h - \cos h}{3h^2} = \lim_{h \rightarrow 0} \frac{-\sin h}{3h}.$$

Once more, L'Hôpital applies and yields

$$\lim_{h \rightarrow 0} \frac{-\sin h}{3h} = \lim_{h \rightarrow 0} \frac{-\cos h}{3} = -\frac{1}{3}.$$

In conclusion, we have $f''(0) = -\frac{1}{3}$.

3. Apply integration by parts with $u = f(t)$ and $dv = \sin(\lambda t) dt$ to obtain

$$\begin{aligned} \int_a^b f(t) \sin(\lambda t) dt &= -\frac{1}{\lambda} f(t) \cos(\lambda t) \Big|_{t=a}^{t=b} + \frac{1}{\lambda} \int_a^b f'(t) \cos(\lambda t) dt \\ &= \frac{1}{\lambda} \left(f(a) \cos(\lambda a) - f(b) \cos(\lambda b) + \int_a^b f'(t) \cos(\lambda t) dt \right), \end{aligned}$$

which is valid because $f'(t)$ and hence $f'(t) \cos(\lambda t)$ is an integrable function of t . We claim that the expression in parentheses is a bounded function of λ . For the first two terms, note that

$$|f(a) \cos(\lambda a) - f(b) \cos(\lambda b)| \leq |f(a) \cos(\lambda a)| + |f(b) \cos(\lambda b)| \leq |f(a)| + |f(b)|.$$

As for the integral, we have

$$\left| \int_a^b f'(t) \cos(\lambda t) dt \right| \leq \int_a^b |f'(t) \cos(\lambda t)| dt \leq \int_a^b |f'(t)| dt,$$

where the first inequality used #2 from the first midterm exam, and the second inequality used #3(a) from Problem Set 2. Combining these two estimates implies that

$$\left| \frac{1}{\lambda} \left(f(a) \cos(\lambda a) - f(b) \cos(\lambda b) + \int_a^b f'(t) \cos(\lambda t) dt \right) \right| \leq \frac{M}{\lambda}$$

for some constant $M \geq 0$, whence

$$\lim_{\lambda \rightarrow \infty} \frac{1}{\lambda} \left(f(a) \cos(\lambda a) - f(b) \cos(\lambda b) + \int_a^b f'(t) \cos(\lambda t) dt \right) = 0$$

by the squeeze theorem.